Exercises

Problem 9.1: Perfect secrecy

A condition for perfect secrecy in a secure communication scheme is that an eavesdropper can learn nothing about the message given the cipher (the encoded message). That is, the probability for any given message is the same whether or not the cipher is known

$$P(m|c) = P(m) \quad \forall\, m, c. \tag{9.1}$$

(a) Show that this is equivalent to the requirement that every message leads to every possible cipher with equal probability

$$P(c|m) = P(c) \quad \forall\, m, c.$$

Problem 9.2: Six-state BB84

Recall that in the standard version of the BB84 QKD protocol, one party (Alice) chooses randomly between the x- and y-basis and encodes random bits into eigenstates of the Pauli operators $\hat{\sigma}_x$ and $\hat{\sigma}_z$. The states are transmitted to a second party (Bob), who chooses randomly between measuring in the $\hat{\sigma}_x$ or $\hat{\sigma}_z$ eigenbases. A man-in-the-middle (intercept-resend) attack by an eavesdropper (Eve) will introduce an error rate of 25% (so the protocol is secure against intercept-resend attacks only for error rates below 24%).

(a) Suggest a generalisation of the BB84 protocol where Alice uses all six eigenstates of the Pauli operators $\hat{\sigma}_x$, $\hat{\sigma}_y$, and $\hat{\sigma}_z$, and Bob makes measurements of $\hat{\sigma}_x$, $\hat{\sigma}_y$, and $\hat{\sigma}_z$.

(b) Discuss the tolerance of the new protocol to errors and to eavesdropping using intercept-resend attacks. Compare to standard BB84.

Problem 9.3: Coherent-state prepare-and-measure

The *conditional Shannon entropy* of Y given X, for a pair of (classical) random variables X, Y is defined as

$$H(Y|X) = \sum_x p(x) H(Y|X = x) \tag{9.2}$$

$$= -\sum_x p(x) \sum_y p(y|x) \log_2 p(y|x) \tag{9.3}$$

$$= -\sum_{x,y} p(x,y) \log_2 p(y|x). \tag{9.4}$$

It measures the additional information gained upon learning about Y when already knowing X. In particular,

$$H(X,Y) = H(X) + H(Y|X), \tag{9.5}$$

where $H(X,Y)$ is the Shannon entropy of the joint distribution of X and Y.

The *mutual information* is a measure of correlations between random variables. It is defined as

$$I(X:Y) = H(X) + H(Y) - H(X,Y). \tag{9.6}$$